

Lanthanides

Normally shown as a strip of elements along the bottom of the periodic table (along with the actinides), this series of metals is found in the sixth period (row) between Group 2 and the transition metals. The lanthanides are named after lanthanum, the first element in this series.

Swedish discovery

The lanthanides have very similar chemical and physical properties, and that made it hard for chemists to tell them apart. The discovery of the mineral gadolinite in 1787, in the village of Ytterby, Sweden, was a breakthrough. Gadolinite can be broken down into various lanthanides, and after a century of studying it, nine lanthanides were eventually found. Four of these – erbium, terbium, ytterbium, and yttrium – were named after Ytterby.



**Black crystals of
gadolinite in rock**

Rare earths

Because they were found combined together in various minerals in Earth's crust, the lanthanides were called “rare earth metals” and were thought to be hard to find. Despite the name, lanthanides are quite common. They are found in low concentrations through all kinds of rock, and are seldom seen in large quantities in one place.



*This mineral contains traces
of almost all lanthanides.*

**Chunk of monazite, an
ore of many lanthanides**